United States Patent	12412436
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Hori; Takashige et al.

Information processing device, information processing method and program

Abstract

An information processing device includes: an accident/near miss detecting unit that detects a first vehicle of a plurality of vehicles based on vehicle information acquired by each of the vehicles and including position information and date-and-time information corresponding to time when the vehicle information is acquired by the vehicles, the first vehicle being involved in a near miss or accident; a surrounding vehicle extracting unit that extracts surrounding vehicles of the vehicles and the date-and-time information corresponding to the position information, the surrounding vehicles being positioned at an area surrounding the first vehicle at time of the near miss or accident; and an object vehicle specifying unit that specifies a second vehicle of the surrounding vehicles, the second vehicle being a vehicle in which an inadequate driving operation causing the near miss or accident of the first vehicle is performed.

Inventors: **Hori; Takashige (Aisai, JP), Sasagawa; Naoto (Nishio, JP)**

Applicant: **TOYOTA JIDOSHA KABUSHIKI KAISHA** (Toyota, JP)

Family ID: **71610500**

Assignee: **TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)**

Appl. No.: **18/350214**

Filed: **July 11, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230351823 A1	Nov. 02, 2023

Foreign Application Priority Data

JP	2019-009755	Jan. 23, 2019

Related U.S. Application Data

continuation parent-doc US 15929205 20200116 US 11734967 child-doc US 18350214

Publication Classification

Int. Cl.: **G07C5/08** (20060101); **B60W30/09** (20120101); **B60W30/095** (20120101); **B60W50/14** (20200101); **G07C5/00** (20060101)

U.S. Cl.:

CPC **G07C5/0833** (20130101); **B60W30/09** (20130101); **B60W30/0956** (20130101); **B60W50/14** (20130101); **G07C5/008** (20130101); **G07C5/0825** (20130101);

Field of Classification Search

CPC: G07C (5/0833); G07C (5/008); G07C (5/0825); G07C (5/085); B60W (30/09); B60W (30/0956); B60W (50/14); G06Q (50/40); G08G (1/0129); G08G (1/0137); G08G (1/166)

USPC: 701/301

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8825277	12/2013	McClellan	701/34.2	G06Q 10/06
9679487	12/2016	Hayward	N/A	G08G 1/096827
11734967	12/2022	Hori	701/301	G07C 5/085
2002/0198632	12/2001	Breed	701/472	G01S 17/86
2007/0135989	12/2006	Hengst	701/117	G08G 1/163
2007/0241932	12/2006	Otero	340/901	G08G 1/096775
2008/0167819	12/2007	Breed	701/300	G08G 1/161
2012/0126997	12/2011	Bensoussan	340/436	B60Q 9/008
2013/0181850	12/2012	Emura	N/A	N/A
2013/0279491	12/2012	Rubin	370/347	G08G 1/162
2013/0338877	12/2012	Straus	701/41	G08G 1/161
2014/0191858	12/2013	Morgan	N/A	N/A
2015/0199162	12/2014	Platz	N/A	N/A
2016/0196098	12/2015	Roth	N/A	N/A
2016/0232790	12/2015	Massey	N/A	G08G 1/096716
2017/0210323	12/2016	Cordova	N/A	G08G 1/0129
2019/0156671	12/2018	Oe	N/A	G01C 21/3461
2020/0231147	12/2019	Hori	N/A	B60W 30/09

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013-33324	12/2012	JP	N/A
2014-10496	12/2013	JP	N/A
2016-071492	12/2015	JP	N/A

2017-182490	12/2016	JP	N/A

OTHER PUBLICATIONS

Office Action mailed Jul. 7, 2021, in U.S. Appl. No. 15/929,205. cited by applicant
Final Office Action mailed May 25, 2002, in U.S. Appl. No. 15/929,205. cited by applicant
Notice of Allowance mailed Apr. 12, 2023, in U.S. Appl. No. 15/929,205. cited by applicant

---

Primary Examiner: Paige; Tyler D

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

INCORPORATION BY REFERENCE (1) This application is a continuation of U.S. application Ser. No. 15/929,205, filed Jan. 16, 2020, which claims the benefit of priority of Japanese Patent Application No. 2019-009755 filed on Jan. 23, 2019, the contents of each of which are incorporated herein by reference in their entirety.

BACKGROUND
1. Technical Field
(1) The disclosure relates to an information processing device and the like.
2. Description of Related Art
(2) For example, there is known a technology of performing determination of occurrence of an accident or near miss of a vehicle, analysis of an occurrence cause, and the like, from common points between vehicle information collected from the vehicle and vehicle information accumulated about accidents and near misses that occurred in the past (see Japanese Patent Application Publication No. 2016-071492, for example).
SUMMARY
(3) By an inadequate driving operation (for example, a steering operation corresponding to a sudden lane change or a forceful cutting into vehicles arrayed in a front-rear direction) in a certain vehicle, a different vehicle can be involved in a near miss or an accident. Therefore, from the standpoint of safety and the like, it is desirable to specify the vehicle in which such an inadequate driving operation has been performed and to perform some kind of reaction to a driver or the like of the vehicle.
(4) However, in the above-described technology, although it is possible to detect a near miss or an accident of a certain vehicle, it is not possible to specify the vehicle in which the inadequate driving operation causing the near miss or the accident has been performed.
(5) Hence, in view of the above problem, the disclosure has an object to provide an information processing device and the like that make it possible to specify the vehicle in which the inadequate driving operation has been performed.
(6) For achieving the above object, an embodiment of the disclosure provides an information processing device including: a detecting unit that detects a first vehicle of a plurality of vehicles based on predetermined vehicle information and date-and-time information, the first vehicle being involved in a near miss or an accident, the predetermined vehicle information being acquired by each of the plurality of vehicles and including position information, the date-and-time information corresponding to time when the vehicle information is acquired by the plurality of vehicles; an extracting unit that extracts surrounding vehicles of the plurality of vehicles based on the position information and the date-and-time information, in a case where the first vehicle is detected by the detecting unit, the surrounding vehicles being positioned at an area surrounding the first vehicle

when the first vehicle is involved in the near miss or the accident; and a specifying unit that specifies a second vehicle of the surrounding vehicles based on the vehicle information about the surrounding vehicles, the second vehicle being a vehicle in which a predetermined driving operation is performed, the predetermined driving operation causing the near miss or the accident of the first vehicle.

(7) With the embodiment, the information processing device can know a motion state of the vehicle, a driving operation state by the driver, and the like, from the vehicle information acquired by each vehicle, and thereby, can detect occurrence of the near miss (for example, a sudden braking) or accident (for example, a collision) of the vehicle. Further, the information processing device can extract the surrounding vehicles positioned at the area surrounding the first vehicle when the first vehicle is involved in the near miss or the accident, from the position information (specifically, the position information about the first vehicle and the position information about the other vehicles) about each vehicle and the date-and-time information corresponding to the time when the vehicle information is acquired. Therefore, the information processing device can know the motion states of the surrounding vehicles, the driving operation states by the driver, and the like, from the extracted vehicle information about the surrounding vehicles, and thereby, can specify the second vehicle in which an inadequate driving operation corresponding to a cause of the near miss or accident of the first vehicle has been performed, from the extracted surrounding vehicles.

(8) In the above embodiment, the information processing device may further include a notifying unit that gives a notice of a caution to a user of the second vehicle specified by the specifying unit, through a user terminal.

(9) With the embodiment, the information processing device can perform the caution relevant to the inadequate driving operation causing the near miss or accident of the first vehicle, to the user of the second vehicle in which the inadequate driving operation has been performed.

(10) In the above-described embodiment, in a case where the second vehicle is traveling, the notifying unit may give an auditory notice as the notice of the caution to the user that rides in the second vehicle, through an in-vehicle device of the second vehicle or a mobile terminal possessed by the user, each of the in-vehicle device and the mobile terminal being the user terminal, and in a case where the second vehicle is at a standstill, the notifying unit may give a visual notice as the notice of the caution to the user, through the in-vehicle device or the mobile terminal.

(11) With the embodiment, the information processing device can give the auditory notice to the user of the second vehicle during traveling, so as not to obstruct the driving operation, and can give the visual notice to the user of the second vehicle at a standstill, so as to perform a more specific caution.

(12) In the above-described embodiment, in a case where the second vehicle is in a stop state, the notifying unit may give the notice of the caution to the user of the second vehicle, through the user terminal different from an in-vehicle device of the second vehicle.

(13) With the embodiment, the information processing device can perform the caution relevant to the inadequate driving operation to the user of the second vehicle, at such a timing that the driving operation of the second vehicle is not obstructed.

(14) In the above-described embodiment, the detecting unit may successively perform the detection of the first vehicle in response to movement of the plurality of vehicles, the extracting unit and the specifying unit may respectively perform the extraction of the surrounding vehicles and the specification of the second vehicle, immediately in response to the detection of the first vehicle by the detecting unit, and in response to the specification of the second vehicle by the specifying unit, the notifying unit may give an auditory notice as the notice of the caution to the user of the second vehicle that rides in the second vehicle, through an in-vehicle device of the second vehicle or a mobile terminal possessed by the user, each of the in-vehicle device and the mobile terminal being the user terminal, at a first timing that is a relatively early timing after the first vehicle is involved

in the near miss or the accident, and then may give a visual notice as the notice of the caution to the user that rides in the second vehicle, through the in-vehicle device or the mobile terminal, at a second timing that is a relatively late timing at which the second vehicle is at a standstill.
(15) With the embodiment, the information processing device can perform a simplified caution to the user (driver) of the second vehicle, shortly after the accident or the near miss, even during traveling, and can perform a specific caution later, in the state where the second vehicle is at a standstill.
(16) In the above-described embodiment, in a case where it is determined that the user is not looking at a content of the visual notice at the second timing, the notifying unit may give the visual notice to the user through the user terminal different from the in-vehicle device of the second vehicle, when the second vehicle gets into a stop state.
(17) With the embodiment, even in the case where the user is not looking at the content of the notice during the driving of the second vehicle, the information processing device can perform again the specific caution to the user of the second vehicle, through the user terminal (for example, a smartphone) other than the in-vehicle device, after the stop of the second vehicle.
(18) Other embodiments of the disclosure can be realized as an information processing method and a program.
(19) The above-described embodiments can provide the information processing device and the like that make it possible to specify the vehicle in which the inadequate driving operation has been performed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like numerals denote like elements, and wherein:
(2) FIG. **1** is a schematic diagram showing an example of a configuration of a vehicle monitoring system;
(3) FIG. **2**A is a diagram showing an example of a hardware configuration of a vehicle;
(4) FIG. **2**B is a diagram showing an example of a hardware configuration of a vehicle monitoring server;
(5) FIG. **3** is a functional block diagram showing an example of a functional configuration of the vehicle monitoring system;
(6) FIG. **4**A is a diagram showing a first example of a caution notice process by the vehicle monitoring server;
(7) FIG. **4**B is a diagram showing an example of a simplified caution notice screen that is displayed on a display of a vehicle;
(8) FIG. **4**C is a diagram showing an example of a detailed caution notice screen that is displayed on the display of the vehicle;
(9) FIG. **5**A is a flowchart schematically showing a second example of the caution notice process by the vehicle monitoring server;
(10) FIG. **5**B is a diagram showing an example of a simplified caution notice screen that is displayed on a display device of a user terminal;
(11) FIG. **5**C is a diagram showing an example of a detailed caution notice screen that is displayed on the display device of the user terminal; and
(12) FIG. **6** is a flowchart schematically showing a third example of the caution notice process by the vehicle monitoring server.
DETAILED DESCRIPTION OF EMBODIMENTS

(13) Hereinafter, an embodiment of the disclosure will be described with reference to the drawings.
(14) Outline of Vehicle Monitoring System
(15) First, an outline of a vehicle monitoring system **1** according to the embodiment will be described with reference to FIG. **1**.
(16) The vehicle monitoring system **1** includes a plurality of vehicles **10**, a vehicle monitoring server **20**, and a plurality of user terminals **40**.
(17) The vehicle monitoring system **1** monitors various abnormalities of the plurality of vehicles **10**, with the vehicle monitoring server **20**. For example, with the vehicle monitoring server **20**, the vehicle monitoring system **1** detects an accident or near miss of a vehicle **10**, and in the case where a cause of the detected accident or near miss is an inadequate driving operation in a different vehicle **10**, the vehicle monitoring system **1** gives a caution relevant to the inadequate driving operation, to a user (driver) of the different vehicle **10**. Here, examples of the inadequate driving operation include a steering operation corresponding to a sudden lane change or a forceful cutting into vehicles arrayed in a front-rear direction, and a braking operation corresponding to an unnecessary sudden braking.
(18) For example, the vehicle **10** is communicably connected with the vehicle monitoring server **20**, through a communication network NW**1** that can include a mobile communication network using base stations as terminals, a satellite communication network using overhead communication satellites, and an internet network. The vehicle **10** acquires a prescribed kind of dynamic information (hereinafter, referred to as “vehicle information”) related to the vehicle **10** (own vehicle), and uploads (sends) the acquired vehicle information to the vehicle monitoring server **20**, in response to a command from the vehicle monitoring server **20** or automatically at a prescribed timing. For example, the vehicle information includes information (hereinafter, referred to as “vehicle state information”) relevant to various states of the vehicle **10** such as a position state of the vehicle **10**, a motion state of the vehicle **10**, an operation state of the vehicle **10** by a driver or the like, and a control state of the vehicle **10**. Here, the vehicle state information includes position information about the vehicle **10**. Further, the vehicle state information includes information (hereinafter, referred to as “vehicle motion state information”) relevant to the motion state of the vehicle **10**. Further, the vehicle state information may include information (hereinafter, referred to as “driving operation information”) relevant to the driving operation of the vehicle **10**. The “driving operation” means a driver's action to actively operate functions of the vehicle **10**, as exemplified by “running”, “turning” and “stopping”, and specifically includes an operation of an accelerator pedal or alternative means for the accelerator pedal, an operation of a steering wheel or alternative means for the steering wheel, and an operation of a brake pedal or alternative means for the brake pedal. Further, for example, the vehicle information may include information relevant to an environment state of the periphery of the vehicle **10**, as exemplified by the ambient temperature of the periphery of the vehicle **10** that is obtained by a temperature sensor, a rainfall state (specifically, existence of a raindrop and the amount of the raindrop) of the periphery of the vehicle **10** that is obtained by a rain sensor, and a traffic state of the periphery of the vehicle **10** that is obtained by an in-vehicle camera.
(19) The vehicle monitoring server **20** (an example of the information processing device) is communicably connected with each of the plurality of vehicles **10**, through the communication network NW**1**. For example, the vehicle monitoring server **20** is communicably connected with a plurality of user terminals **40**, through a communication network NW**2** that can include a mobile communication network using base stations as terminals, a satellite communication network using overhead communication satellites, and an internet network.
(20) The vehicle monitoring server **20** receives the vehicle information that is sent from each of the plurality of vehicles **10**, and accumulates the received vehicle information for each vehicle **10**.
(21) Further, for each of the plurality of vehicles **10**, the vehicle monitoring server **20** detects occurrence of the accident or near miss of the vehicle **10**, based on the vehicle information received

from the vehicle **10**. Then, in the case where the cause of the detected accident or near miss is an inadequate operation in a different vehicle **10**, the vehicle monitoring server **20** gives the caution relevant to the inadequate driving operation, to the user (driver) of the different vehicle **10**, through an in-vehicle device (for example, a later-described navigation device **15**) of the different vehicle **10** or a user terminal **40** that is used by the user (driver) of the different vehicle **10**.
(22) The user terminal **40** is used by the user of the vehicle **10**. For example, the user terminal **40** is a mobile terminal such as a smartphone, a mobile phone, a tablet terminal and a laptop computer. Further, for example, the user terminal **40** may be a stationary terminal such as a desktop computer. Further, for example, the user terminal **40** may be an in-vehicle device (in-vehicle terminal) equipped in the vehicle **10**, as exemplified by the later-described navigation device **15** or an audio device.
(23) The user terminal **40** is communicably connected with the vehicle monitoring server **20**, through the communication network NW**2**. For example, the user terminal **40** gives a variety of information (for example, caution information relevant to the inadequate driving operation in the vehicle **10** that is used by the user of the user terminal **40**) delivered from the vehicle monitoring server **20** to the user, by visual display or voice.
(24) Configuration of Vehicle Monitoring System
(25) Next, the configuration of the vehicle monitoring system **1** will be described with reference to FIG. **2**A, FIG. **2**B and FIG. **3**, in addition to FIG. **1**.
(26) Each of FIG. **2**A and FIG. **2**B is a diagram showing an example of a hardware configuration of the vehicle monitoring system **1**. Specifically, FIG. **2**A is a diagram showing an example of a hardware configuration of the vehicle **10**, and FIG. **2**B is a diagram showing an example of a hardware configuration of the vehicle monitoring server **20**. FIG. **3** is a functional block diagram showing an example of a functional configuration of the vehicle monitoring system **1**.
(27) A hardware configuration of the user terminal **40** is nearly the same as the hardware configuration of the vehicle monitoring server **20**. Therefore, descriptions will be made below, with reference to FIG. **2**B, and illustration of the hardware configuration of the user terminal **40** will be omitted. Hereinafter, in the description of the user terminal **40**, reference characters "**21**", "**21**A", "**22**", "**23**", "**24**", "**25**", "**26**", "**27**" and "B**2**" in FIG. **2**B are replaced with "**41**", "**41**A", "**42**", "**43**", "**44**", "**45**", "**46**", "**47**" and "B**4**", respectively.
Configuration of Vehicle
(28) As shown in FIG. **2**A, the vehicle **10** includes an ECU **11**, a global navigation satellite system (GNSS) module **12**, a data communication module (DCM) **13**, a sensor **14**, the navigation device **15**, and a speaker **16**.
(29) The ECU **11** is an electronic control unit that performs various controls relevant to the vehicle **10**. Functions of the ECU **11** may be realized by arbitrary hardware or combinations of hardware and software. For example, the ECU **11** may be constituted mainly by a microcomputer including an auxiliary storage device **11**A, a memory device **11**B, a central processing unit (CPU) **11**C, an interface device **11**D and the like, which are connected with each other by a bus B**1**.
(30) For example, programs to realize various programs of the ECU **11** are provided by a dedicated tool that is connected with a predetermined connector (for example, a data link coupler (DLC)) for external connection by a detachable cable. The connector is joined to an in-vehicle network such as a controller area network (CAN) of the vehicle **10**. In response to a predetermined operation of the dedicated tool, the programs are installed in the auxiliary storage device **11**A of the ECU **11** from the dedicated tool through the cable, the connector and the in-vehicle network. Further, the programs may be installed in the auxiliary storage device **11**A by being downloaded from another computer (for example, the vehicle monitoring server **20**) through the communication network NW**1**.
(31) The auxiliary storage device **11**A, which is non-volatile storage means, contains the installed programs, and contains necessary files, necessary data and the like. For example, the auxiliary

storage device **11**A is a hard disk drive (HDD), a flash memory or the like.
(32) When an activation instruction for a program is given, the memory device **11**B reads the program from the auxiliary storage device **11**A, and contains the program.
(33) The CPU **11**C executes the programs contained in the memory device **11**B, and realizes various functions of the ECU **11** in accordance with the programs.
(34) The interface device **11**D, for example, is used as an interface for connection with the in-vehicle network and one-to-one connection with various sensors, various actuators and the like. The interface device **11**D can include different kinds of interface devices, depending on objects to be connected.
(35) The GNSS module **12** receives satellite signals sent from three or more satellites, preferably, four or more satellites over the vehicle **10**, and thereby, measures the position of the vehicle **10** (own vehicle) equipped with the GNSS module **12**. Positioning information of the GNSS module **12**, that is, position information about the vehicle **10** is taken in the DCM **13**, through a one-to-one communication line or the in-vehicle network, for example. The positioning information of the GNSS module **12** may be taken in the ECU **11** from the DCM **13** through the in-vehicle network or the like, for example.
(36) The DCM **13**, which is connected with the communication network NW**1** in the exterior of the vehicle **10**, is an exemplary communication device for communicating with an external device including the vehicle monitoring server **20** through the communication network NW**1**. The DCM **13** sends and receives various signals (for example, information signals, control signals and the like) to and from the vehicle monitoring server **20**. The DCM **13** is communicably connected with the ECU **11** through the one-to-one communication line or the in-vehicle network such as the CAN. In response to requests from the ECU **11**, the DCM **13** sends various signals to the exterior of the vehicle **10** (own vehicle), or outputs signals received from the exterior of the vehicle **10**, to the ECU **11**.
(37) The sensor **14** outputs detection information corresponding to at least one of the driving operation information and motion state information about the vehicle **10**. For example, the sensor **14** can include an accelerator pedal sensor that detects an operation state of the accelerator pedal of the vehicle **10**, a brake pedal sensor that detects an operation state of the brake pedal, and a steering wheel sensor that detects an operation state of the steering wheel. Further, for example, the sensor **14** can include an acceleration sensor that detects the acceleration of the vehicle **10**, and an angular velocity sensor (gyro sensor) that detects the angular velocity of the vehicle **10**. The detection information output from the sensor **14** is taken in the ECU **11**, through a one-to-one communication line or the in-vehicle network such as the CAN.
(38) The navigation device **15** (an example of the user terminal or the in-vehicle device), which is equipped in a vehicle cabin of the vehicle **10**, searches a route from the current place of the vehicle **10** to a destination to be set, and guides the driver of the vehicle **10** on a movement path along the route as the search result. The navigation device **15** includes a head unit **15**A and a display **15**B.
(39) The head unit **15**A is a principal unit that realizes the route search function and the route guide function. For example, the head unit **15**A performs the route search, by a known method, based on position information corresponding to the current place of the vehicle **10** that is acquired by the GNSS module **12** and destination information that is input by an operation from the user. Further, the head unit **15**A controls the display **15**B and the speaker **16**, to perform the route guide.
(40) The route search function may be realized by an external navigation server. In this case, the head unit **15**A may send information for the route search, as exemplified by the destination information input from the user, to the external navigation server, through the DCM **13**, and may perform the route guide, using the display **15**B and the speaker **16**, based on information that is relevant to the route guide and that is sent back from the navigation server.
(41) The display **15**B displays information (for example, a map image, route information for the guide, and the like) relevant to the route guide, under control of the head unit **15**A. Further, the

display **15**B displays various information images, under control of the head unit **15**A based on a request from the ECU **11**.
(42) The speaker **16**, which is equipped in the vehicle cabin of the vehicle **10**, outputs sound under control of the head unit **15**A.
(43) As shown in FIG. **3**, for example, the ECU **11** includes a vehicle information sending unit **111** and a notifying unit **112**, as functional units that are realized when the CPU **11**C executes one or more programs installed in the auxiliary storage device **11**A.
(44) For example, in a predetermined cycle (for example, in a cycle of several seconds to several tens of seconds), the vehicle information sending unit **111** acquires a prescribed kind of vehicle information or causes the DCM **13** to acquire the vehicle information, and sends the vehicle information to the vehicle monitoring server **20** through the DCM **13**. The vehicle information to be sent to the vehicle monitoring server **20** includes the position information about the vehicle **10** that is obtained by the GNSS module **12**. Further, the vehicle information to be sent to the vehicle monitoring server **20** includes the detection information of the sensor **14** that corresponds to at least one of the driving operation information and the motion state information. Specifically, to the vehicle monitoring server **20**, the vehicle information sending unit **111** may send a signal including identification information (for example, the vehicle identification number (VIN) of the vehicle **10** or a vehicle identifier (ID) prescribed for each of the plurality of vehicles **10**; hereinafter, referred to as “vehicle identification information”) for specifying the vehicle **10** that is a sending source, information (for example, time stamp; hereinafter, referred to as “acquisition date-and-time information”) relevant to the date and time of the acquisition of the vehicle information, and the vehicle information. Thereby, the vehicle monitoring server **20** can identify (specify) the vehicle **10** that is the sending source of the signal including the vehicle information, or can specify the date and time (acquisition timing) of the acquisition of the vehicle information, and the like.
(45) The function of the vehicle information sending unit **111** may be transferred to the DCM **13**.
(46) The notifying unit **112** gives a variety of information (for example, the above-described caution information relevant to the inadequate driving operation) received from the vehicle monitoring server **20**, to the user (driver) of the vehicle **10**, through the display **15**B or the speaker **16**. Specifically, the notifying unit **112** controls the display **15**B and the speaker **16**, through the head unit **15**A. Thereby, the notifying unit **112** displays an information image on the display **15**B, and outputs voice information from the speaker **16**. Details will be described later (see FIG. **4**B).
(47) Configuration of Vehicle Monitoring Server
(48) Functions of the vehicle monitoring server **20** may be realized by arbitrary hardware or combinations of hardware and software. As shown in FIG. **2**B, for example, the vehicle monitoring server **20** includes a drive device **21**, an auxiliary storage device **22**, a memory device **23**, a CPU **24**, an interface device **25**, a display device **26**, and input device **27**, which are connected with each other by a bus B**2**.
(49) For example, programs to realize various functions of the vehicle monitoring server **20** are by a portable recording medium **21**A such as a compact disc read only memory (CD-ROM), a digital versatile disc read only memory (DVD-ROM) or a universal serial bus (USB) memory. When the recording medium **21**A in which the programs are recorded are set in the drive device **21**, the programs are installed in the auxiliary storage device **22** from the recording medium **21**A through the drive device **21**. Further, the programs may be downloaded from another computer through the communication network, and may be installed in the auxiliary storage device **22**.
(50) The auxiliary storage device **22** contains the installed various programs, and contains necessary files, necessary data and the like.
(51) When an activation instruction for a program is given, the memory device **23** reads the program from the auxiliary storage device **22**, and contains the program.
(52) The CPU **24** executes the various programs contained in the memory device **23**, and realizes various functions about the vehicle monitoring server **20** in accordance with the programs.

(53) The interface device **25** is used as an interface for connection with communication networks (for example, the communication networks NW**1**, NW**2**).
(54) For example, the display device **26** displays a graphical user interface (GUI) in accordance with programs that are executed by the CPU **24**.
(55) The input device **27** is used when an operator, an administrator or the like of the vehicle monitoring server **20** inputs various operation instructions relevant to the vehicle monitoring server **20**.
(56) As shown in FIG. **3**, for example, the vehicle monitoring server **20** includes a vehicle information acquiring unit **201**, an accident/near miss detecting unit **203**, a surrounding vehicle extracting unit **204**, an object vehicle specifying unit **205**, and a notice control unit **206**, as functional units that are realized when the CPU **24** executes one or more programs installed in the auxiliary storage device **22**. The vehicle monitoring server **20** uses a vehicle information storage unit **202** and the like. For example, the vehicle information storage unit **202** can be realized using the auxiliary storage device **22** or an external storage device or the like that is communicably connected with the vehicle monitoring server **20**.
(57) The vehicle information acquiring unit **201** acquires the vehicle information included in the signal received from each of the plurality of vehicles **10**, and stores (accumulates) the vehicle information in the vehicle information storage unit **202**. Specifically, the vehicle information acquiring unit **201** stores the vehicle information received from the vehicle **10**, in the vehicle information storage unit **202**, as a record associated with the corresponding vehicle identification information and acquisition date-and-time information.
(58) As described above, the vehicle information received from the vehicle **10** is stored in the vehicle information storage unit **202**. Specifically, the vehicle information storage unit **202** may hold a record group (that is, a database) of the vehicle information acquired by the plurality of vehicles **10**, by accumulating records including the vehicle identification information, the acquisition data-and-time information and the vehicle information. Further, the vehicle information storage unit **202** may be provided with a vehicle information storage unit dedicated for each of the plurality of vehicles **10**, and may hold a history of records including the acquisition date-and-time information and the vehicle information for each vehicle **10**, that is, a record group, in the corresponding vehicle information storage unit.
(59) The accident/near miss detecting unit **203** (an example of the detecting unit) detects the vehicle **10** involved in an accident or a near miss, based on the vehicle information for each of the plurality of vehicles **10**, which is information stored in the vehicle information storage unit **202**. Examples of the accident to be detected include a collision with a different vehicle **10**, a collision with a feature other than the vehicle **10**, and an overturn (rollover) of the vehicle **10**. Specifically, the accident/near miss detecting unit **203** is included in the vehicle information. Examples of the near miss to be detected include a sudden braking or sudden steering of the vehicle **10**. The accident/near miss detecting unit **203** detects the vehicle **10** involved in the accident or the near miss, based on the motion state information (for example, information relevant to the acceleration and angular acceleration of the vehicle **10**) and driving operation information about the vehicle **10**. The accident/near miss detecting unit **203** may perform the detection of the vehicle **10** involved in the accident or the near miss, in real time, depending on a movement situation of the vehicle **10**, or may perform the detection of the vehicle **10** involved in the accident or the near miss, as a batch process, at a decided cyclic timing. The same goes for the surrounding vehicle extracting unit **204**, the object vehicle specifying unit **205** and the notice control unit **206**, which operate when the accident/near miss detecting unit **203** detects the vehicle **10** involved in the accident or the near miss. Hereinafter, in the embodiment, descriptions will be made, assuming that the accident/near miss detecting unit **203** performs the detection of the vehicle **10** involved in the accident or the near miss in real time and the functions of the surrounding vehicle extracting unit **204**, the object vehicle specifying unit **205** and the notice control unit **206** operate simultaneously in response to the

detection in real time.
(60) In the case where the accident/near miss detecting unit **203** has detected the vehicle **10** involved in the accident or the near miss, the surrounding vehicle extracting unit **204** (an example of the extracting unit) extracts vehicles **10** (hereinafter, referred to as “surrounding vehicles”) that are positioned at an area (for example, an area within several meters to several tens of meters) surrounding the vehicle **10** (hereinafter, referred to as “detected vehicle”) when the detected vehicle **10** is involved in the accident or the near miss, from the plurality of vehicles **10** other than the detected vehicle **10**. Specifically, the surrounding vehicle extracting unit **204** extracts the surrounding vehicles from the plurality of vehicles **10** other than the detected vehicle, based on the position information about each of the plurality of vehicles **10** other than the detected vehicle when the detected vehicle is involved in the accident or the near miss. On this occasion, based on the acquisition date-and-time information of the record group in the vehicle information storage unit **202**, the surrounding vehicle extracting unit **204** can acquire the record corresponding to the vehicle information (position information) when the detected vehicle is involved in the accident or the near miss, from the record group in the vehicle information storage unit **202**.
(61) The object vehicle specifying unit **205** (an example of the specifying unit) specifies a vehicle **10** (hereinafter, referred to as an “object vehicle”) in which an inadequate driving operation causing the accident or near miss of the detected vehicle has been performed, from the surrounding vehicles extracted by the surrounding vehicle extracting unit **204**. Naturally, there can be an accident or near miss of the detected vehicle that is caused by the detected vehicle itself, and therefore, in some cases, the object vehicle specifying unit **205** does not specify (cannot specify) the object vehicle.
(62) In the case where the object vehicle specifying unit **205** has specified the object vehicle, the notice control unit **206** (an example of the notifying unit) sends, to the object vehicle (vehicle **10**) or the user terminal **40**, the caution information for giving a notice of a caution relevant to the inadequate driving operation to the user (driver) of the object vehicle. Thereby, the notice control unit **206** can perform the caution to the user of the object vehicle in which the inadequate driving operation has been performed, through the head unit **15**A of the vehicle **10** or the user terminal **40**. Details will be described later (see FIG. **4**A, FIG. **5**A and FIG. **6**).
(63) The functions of the accident/near miss detecting unit **203**, the surrounding vehicle extracting unit **204**, the object vehicle specifying unit **205** and the notice control unit **206** may be transferred to an external device (for example, another server device; an example of the information processing device) that can communicate with the vehicle monitoring server **20**.
(64) Configuration of User Terminal
(65) Similarly to the vehicle monitoring server **20**, functions of the user terminal **40** may be realized by arbitrary hardware or combinations of hardware and software. As shown in FIG. **2**B, for example, the user terminal **40** includes a drive device **41**, an auxiliary storage device **42**, a memory device **43**, a CPU **44**, an interface device **45**, a display device **46** and an input device **47**, which are connected with each other by a bus B**4**.
(66) A hardware configuration of the user terminal **40** is nearly the same as the hardware configurations of the vehicle monitoring server **20** and the like, and therefore, detailed descriptions will be omitted.
(67) As shown in FIG. **3**, for example, the user terminal **40** includes a notifying unit **401**, as a functional unit that is realized when the CPU **44** executes one or more programs installed in the auxiliary storage device **42**.
(68) The notifying unit **401** gives a variety of information (for example, the above-described caution information relevant to the inadequate driving operation) received from the vehicle monitoring server **20**, to the user (driver) of the vehicle **10**, through the display device **46** or a speaker equipped in the user terminal **40**. Specifically, the notifying unit **401** controls the display device **46** and the speaker. Thereby, the notifying unit **401** displays an information image on the display device **46**, and outputs voice information from the speaker. Details will be described later

(see FIG. **4**C).
(69) Specific Examples of Caution Method Relevant to Inadequate Driving Operation
(70) Next, specific examples of the caution method relevant to the inadequate driving operation by the vehicle monitoring system **1** will be described with reference to FIGS. **4**A to **4**C, FIGS. **5**A to **5**C and FIG. **6**.
(71) First Example of Caution Method Relevant to Inadequate Driving Operation
(72) In a first example, the caution relevant to the inadequate driving operation is performed to the user of the object vehicle, through the in-vehicle device (navigation device **15**) of the object vehicle.
(73) FIG. **4**A to FIG. **4**C are diagrams for describing the first example of the caution method relevant to the inadequate driving operation by the vehicle monitoring server **20**. Specifically, FIG. **4**A is a flowchart schematically showing the first example of a notice process (hereinafter, referred to as a “caution notice process”) for the caution relevant to the inadequate driving operation by the vehicle monitoring server **20**. FIG. **4**B is a diagram showing an example (simplified caution notice screen **410**) of a screen (hereinafter, referred to as a “simplified caution notice screen”) for giving simplified caution information that is displayed on the display **15**B of the vehicle **10** (detected vehicle). FIG. **4**C is a diagram showing an example (detailed caution notice screen **420**) of a screen (hereinafter, referred to as a “detailed caution notice screen”) for giving the caution information that is displayed on the display **15**B of the vehicle **10** (detected vehicle).
(74) The flowchart in FIG. **4**A starts to be executed when the object vehicle specifying unit **205** has specified the object vehicle. The flowchart in FIG. **4**A is forcibly terminated when the object vehicle is stopped, that is, ignition-off (hereinafter, referred to as “IG-OFF”) is performed during execution of a process in or after step S**104**.
(75) As shown in FIG. **4**A, in step S**101**, the notice control unit **206** determines whether the object vehicle is in an ignition-on (hereinafter, referred to as “IG-ON”) state (that is, whether the object vehicle has been activated such that the object vehicle can travel), based on the most recent vehicle information about the object vehicle in the vehicle information storage unit **202**. For example, the notice control unit **206** may perform the determination from the acquisition date-and-time information of the most recent record about the object vehicle that is stored in the vehicle information storage unit **202**. This is because the signal including the vehicle information is not uploaded from the object vehicle in the case of the stop of the object vehicle, that is, in the case of the IG-OFF. In the case where the object vehicle is in the IG-ON state, the notice control unit **206** proceeds to step S**102**, and in the case where the object vehicle is in the IG-OFF state, the notice control unit **206** ends this process.
(76) In step S**102**, the notice control unit **206** determines whether the object vehicle is traveling, based on the most recent history of the position information about the object vehicle that is stored in the vehicle information storage unit **202**. In the case where the object vehicle is traveling, the notice control unit **206** proceeds to step S**104**, and in the case where the object vehicle is at a standstill, the notice control unit **206** proceeds to step S**108**.
(77) In step S**104**, the notice control unit **206** sends, to the object vehicle, caution information for performing a simplified caution to the user (driver) of the object vehicle by an auditory method. Thereby, the ECU **11** (notifying unit **112**) of the object vehicle receives the caution information, and gives a notice (hereinafter, referred to as a “simplified caution notice”) of the simplified caution, by causing the head unit **15**A to output a predetermined warning sound or a predetermined voice from the speaker **16**. That is, the notice control unit **206** causes the notifying unit **112** of the object vehicle to give the simplified caution notice to the user of the object vehicle by the auditory method. For example, the notifying unit **112** of the object vehicle may give the simplified caution notice relevant to the inadequate driving operation, with a voice such as “Your driving caused a sudden braking of another vehicle”. Thereby, the notice control unit **206** can perform a minimal caution by the auditory method, so as not to interfere with the driving of the user (driver) of the

object vehicle during traveling.

(78) In step S**106**, the notice control unit **206** determines whether the object vehicle is continuously at a standstill, based on the most recent vehicle information (for example, the position information, vehicle speed information and the like) about the object vehicle that is stored in the vehicle information storage unit **202**. For example, in the case where a predetermined number of most recent pieces of vehicle information indicate that the object vehicle is at standstill, the notice control unit **206** may determine that the object vehicle is continuously at a standstill. In the case where the object vehicle is continuously at a standstill, the notice control unit **206** proceeds to step S**108**, and in the case where the object vehicle is not continuously at a standstill, the notice control unit **206** waits until the standstill state is continued (that is, the notice control unit **206** repeats the process of step S**106** until the standstill state is continued).

(79) Between step S**104** and step S**106**, a process of determining whether a predetermined time has elapsed since the occurrence of the accident or the near miss (that is, the occurrence of the inadequate driving operation) may be added similarly to step S**304** in FIG. **6** described later.

(80) In step S**108**, the notice control unit **206** sends, to the object vehicle, the caution information for performing the simplified caution to the user (driver) of the object vehicle by the auditory method. Thereby, the ECU **11** (notifying unit **112**) of the object vehicle receives the caution information, and gives the simplified caution notice, by causing the head unit **15**A to display the simplified caution notice screen on the display **15**B. That is, the notice control unit **206** causes the notifying unit **112** of the object vehicle to give the simplified caution notice to the user of the object vehicle by the visual method.

(81) For example, as shown in FIG. **4**B, on the simplified caution notice screen **410**, a pop-up **411** is displayed at an upper edge portion of a display region of the display **15**B, so as to be superimposed on the display content.

(82) The pop-up **411** contains textual information ("INADEQUATE DRIVING PERFORMED") indicating that the user of the object vehicle has performed an inadequate driving operation. Thereby, the user can recognize that there is a possibility that the user has performed the inadequate driving operation. Further, the pop-up **411** contains an opening icon **412** for checking detailed caution information from the vehicle monitoring server **20** (notice control unit **206**). Thereby, the user can display the detailed caution information on the display **15**B, by operating the opening icon **412** through predetermined operation input means (for example, a touch panel mounted on the display **15**B.) attached to the navigation device **15**.

(83) Back to FIG. **4**A, in step S**110**, the notice control unit **206** determines whether an operation (hereinafter, referred to as a "check operation") for checking the detailed caution information (for example, an operation of the opening icon **412** on the simplified caution notice screen **410**) has been performed on the simplified caution notice screen. On this occasion, the notice control unit **206** may perform the determination by making an inquiry to the ECU **11** (notifying unit **112**) of the object vehicle about whether the check operation has been performed, or may perform the determination based on information that is relevant to whether the check operation has been performed and that is automatically uploaded from the notifying unit **112**. In the case where the check operation has been performed, the notice control unit **206** proceeds to step S**112**, and in the case where the check operation has not been performed, the notice control unit **206** waits until the check operation is performed (that is, the notice control unit **206** repeats the process of step S**110**).

(84) In step S**112**, the notice control unit **206** sends, to the object vehicle, the caution information for performing the detailed caution to the user (driver) of the object vehicle by the visual method, and then ends this process. Thereby, the ECU **11** (notifying unit **112**) of the object vehicle receives the caution information, and gives a notice (hereinafter, referred to as a "detailed caution notice") of the detailed caution, by causing the head unit **15**A to display the detailed caution notice screen on the display **15**B. That is, the notice control unit **206** causes the notifying unit **112** of the object vehicle to give the detailed caution notice to the user of the object vehicle by the visual method.

(85) For example, as shown in FIG. **4**C, on the detailed caution notice screen **420**, a pop-up **421** is displayed so as to be superimposed on the display content of the display **15**B and cover a large portion of the display content. The pop-up **421** contains information ("<WARNING>INADEQUATE DRIVING OCCURRED") indicating that an inadequate driving operation has been performed, the date and time ("10/23, ABOUT 13:05") of the occurrence, the content (SUDDEN BRAKING) of a trouble due to the inadequate driving operation, the content (SUDDEN CUTTING-IN) of the inadequate driving operation, and the like. Further, the pop-up **421** contains the place of the occurrence of the accident or the near miss, in other words, a map image **422** corresponding to the place of the occurrence of the inadequate driving operation in the object vehicle. Thereby, the user can know the date and time of the occurrence of the inadequate driving operation by the user, the place of the occurrence, the content of the inadequate driving operation, and the like, and can look back on the user's driving operation at that time. Further, the user can know the content of the trouble due to the inadequate driving operation, and can recognize the generation of the actual trouble due to the driving operation at that time. Therefore, the vehicle monitoring server **20** can perform the detailed caution relevant to the inadequate driving operation to the user of the object vehicle, through the detailed caution notice screen **420**, when the object vehicle is at a standstill.
(86) Between step S**104** and step S**106**, a process of determining whether a predetermined time has elapsed since the occurrence of the accident or the near miss (that is, the occurrence of the inadequate driving operation) may be added similarly to step S**304** in FIG. **6** described later. Thereby, as described later, it is possible to delay the timing when the visual caution is performed to the user, from the occurrence of the inadequate driving operation, to some extent. Further, in the case where the user terminal **40** is a mobile terminal possessed by the user of the object vehicle, a sequence of the caution notice process in FIG. **4**A may be performed such that the caution to the user of the object vehicle is realized through the user terminal **40**. That is, the simplified caution notice by the auditory method in step S**104** may be performed through the speaker of the user terminal **40**, and the simplified caution notice and detailed caution notice by the visual method in step S**108** and step S**112** may be performed through the display device **46** of the user terminal **40**. Further, the sequence of the caution notice process in FIG. **4**A may be autonomously performed by the ECU **11** (notifying unit **112**) of the object vehicle or the user terminal **40** (notifying unit **401**) possessed by the user of the object vehicle, based on the caution information received from the vehicle monitoring server **20** (notice control unit **206**).
(87) Second Example of Caution Method Relevant to Inadequate Driving Operation
(88) In a second example, the caution is performed to the user of the object vehicle, through the user terminal **40** that is used by the user of the object vehicle.
(89) FIG. **5**A to FIG. **5**C are diagrams for describing the second example of the caution method relevant to the inadequate driving operation by the vehicle monitoring server **20**. Specifically, FIG. **5**A is a flowchart schematically showing the second example of the caution notice process relevant to the inadequate driving operation by the vehicle monitoring server **20**. FIG. **5**B is a diagram showing an example (simplified caution notice screen **510**) of a simplified caution notice screen to be displayed on the display device **46** of the user terminal **40** that is used by the user of the object vehicle. FIG. **5**C is a diagram showing an example (detailed caution notice screen **520**) of a detailed caution notice screen to be displayed on the display device **46** of the user terminal **40** that is used by the user of the object vehicle.
(90) For example, the flowchart in FIG. **5**A starts to be executed when the IG-OFF of the object vehicle is performed after the object vehicle is specified by the object vehicle specifying unit **205**.
(91) As shown in FIG. **5**A, in step S**202**, to the user terminal **40** that is used by the user of the object vehicle, the notice control unit **206** sends the caution information for performing the simplified caution to the user of the object vehicle by the visual method. Thereby, the user terminal **40** (notifying unit **401**) that is used by the user of the object vehicle receives the caution

information, and displays the simplified caution notice screen on the display device **46**. That is, the notice control unit **206** causes the notifying unit **401** of the user terminal **40** to give the simplified caution notice to the user of the object vehicle by the visual method. Specifically, association information associated with identification information (for example, a user ID specific to the user; hereinafter, referred to as "user identification information") about the user, the vehicle identification information about the vehicle **10** that is used by the user, and identification information (for example, international mobile equipment identity (IMEI); hereinafter, referred to as "terminal identification information") about the user terminal **40** that is used by the user may be previously registered in the auxiliary storage device **22** of the vehicle monitoring server **20**, or the like. Thereby, the notice control unit **206** can specify the user terminal **40** that is used by the user of the object vehicle, based on the association information, and can send the caution information to the specified user terminal **40**.

(92) For example, as shown in FIG. 5B, on the simplified caution notice screen **510**, a pop-up **511** is displayed at a center portion of a display region of the display device **46**, so as to be superimposed on the display content.

(93) The pop-up **511** contains textual information ("INADEQUATE DRIVING PERFORMED") indicating that the user of the user terminal **40**, that is, the user of the object vehicle has performed an inadequate driving operation. Thereby, the user can recognize that there is a possibility that the user has performed the inadequate driving operation.

(94) The pop-up **511** is also a virtual operation object for checking the detailed caution information from the vehicle monitoring server **20** (notice control unit **206**). Thereby, the user can display the detailed caution information on the display device **46**, for example, by operating the pop-up **511** through the input device **47** such as a touch panel mounted on the display device **46**.

(95) Back to FIG. 5A, in step S**204**, the notice control unit **206** determines whether an operation for checking the detailed caution information (for example, an operation of the pop-up **511** on the simplified caution notice screen **510**) has been performed on the simplified caution notice screen. On this occasion, the notice control unit **206** may perform the determination by making an inquiry to the user terminal **40** (notifying unit **401**) about whether the check operation has been performed, or may perform the determination based on information that is relevant to whether the check operation has been performed and that is automatically uploaded from the notifying unit **401**. The same goes for a determination process in step S**208**. In the case where the check operation has been performed, the notice control unit **206** proceeds to step S**206**, and in the case where the check operation has not been performed, the notice control unit **206** proceeds to step S**208**.

(96) In step S**206**, to the user terminal **40** that is used by the user of the object vehicle, the notice control unit **206** sends the caution information for performing the detailed caution to the user of the object vehicle by the visual method, and then ends this process. Thereby, the user terminal **40** (notifying unit **401**) receives the caution information, and displays the detailed caution notice screen on the display device **46**. That is, the notice control unit **206** causes the notifying unit **401** of the user terminal **40** to give the detailed caution notice to the user of the object vehicle by the visual method. Thereby, the notice control unit **206** can perform the detailed caution relevant to the inadequate driving operation to the user of the object vehicle, at the timing when the user is not driving the object vehicle, that is, without interfering with the driving of the object vehicle by the user.

(97) For example, as shown in FIG. 5C, the detailed caution notice screen **520** contains the date and time ("10/23, 13:05:00-13:05:30") of the occurrence of the inadequate driving operation, the content (SUDDEN BRAKING) of a trouble due to the inadequate driving operation, the content (SUDDEN CUTTING-IN) of the inadequate driving operation, and an advice ("PERFORM LANE CHANGE AFTER CHECKING THAT THERE IS SUFFICIENT DISTANCE AND EXPRESSING YOUR INTENTION TO SURROUNDING VEHICLES WITH TURN SIGNAL") relevant to the driving operation. Further, the detailed caution notice screen **520** contains the place

of the occurrence of the accident or the near miss, in other words, a map image **521** corresponding to the place of the occurrence of the inadequate driving operation in the object vehicle, and a picture **522** of the object vehicle at the time of the inadequate driving operation. The picture **522** is image information of the in-vehicle camera at the time of the occurrence of the accident or the near miss, and is uploaded from the surrounding vehicles. The picture **522** may be played back by an operation through the input device **47**. Thereby, the user can know the date and time of the occurrence of the inadequate driving operation by the user, the place of the occurrence, the content of the inadequate driving operation, the picture at that time, the advice relevant to the driving operation, and the like, can look back on the user's driving operation at that time, and can improve the driving operation in the future. Further, the user can know the content of the trouble due to the inadequate driving operation, and can recognize the generation of the actual trouble due to the driving operation at that time. Therefore, the vehicle monitoring server **20** can perform the detailed caution relevant to the inadequate driving operation to the user of the object vehicle, through the detailed caution notice screen **520** on the user terminal **40**, without interfering with the driving of the object vehicle by the user.
(98) Back to FIG. **5**A, in step S**208**, the notice control unit **206** determines whether the simplified caution notice has been hidden by a user's operation through the input device **47**. In the case where the simplified caution notice has been hidden, the notice control unit **206** proceeds to step S**210**, and in the case where the simplified caution notice has not been hidden, the notice control unit **206** returns to step S**204**.
(99) In step S**210**, the notice control unit **206** determines whether a predetermined time T**1** (for example, ten minutes) has elapsed since the hiding of the simplified caution notice. In the case where the predetermined time T**1** has not elapsed, the notice control unit **206** waits until the elapse of the predetermined time T**1** (that is, the notice control unit **206** repeats the process of step S**210** until the elapse of the predetermined time T**1**), and in the case where the predetermined time T**1** has elapsed, the notice control unit **206** returns to step S**202**, to display the simplified caution notice screen on the display device **46** again. Thereby, even when the user has hidden the simplified caution notice without checking the detailed caution notice, the vehicle monitoring server **20** can display the simplified caution notice screen on the display device **46** again, and can provide an opportunity to perform the detailed caution relevant to the inadequate driving operation to the user of the object vehicle.
(100) A sequence of the caution notice process in FIG. **5**A may be autonomously performed by the user terminal **40** (notifying unit **401**) that is used by the user of the object vehicle, based on the caution information received from the vehicle monitoring server **20** (notice control unit **206**).
(101) Third Example of Caution Method Relevant to Inadequate Driving Operation
(102) In a third example, the caution relevant to the inadequate driving operation is performed to the user of the object vehicle, using both the in-vehicle device of the object vehicle and the user terminal **40** that is used by the user of the object vehicle.
(103) FIG. **6** is a diagram for describing the third example of the caution method relevant to the inadequate driving operation by the vehicle monitoring server **20**. Specifically, FIG. **6** is a flowchart schematically showing the third example of the caution notice process relevant to the inadequate driving operation by the vehicle monitoring server **20**. For example, the flowchart starts to be executed when the object vehicle specifying unit **205** has specified the object vehicle.
(104) As shown in FIG. **6**, in step S**301**, the notice control unit **206** determines whether the object vehicle is in the IG-ON state (that is, whether the object vehicle has been activated such that the object vehicle can travel), based on the most recent vehicle information about the object vehicle in the vehicle information storage unit **202**. In the case where the object vehicle is in the IG-ON state, the notice control unit **206** proceeds to step S**302**, and in the case where the object vehicle is in the IG-OFF state, the notice control unit **206** proceeds to step S**316**.
(105) In step S**302**, the notice control unit **206** sends, to the object vehicle, the caution information

for performing the simplified caution to the user (driver) of the object vehicle by the auditory method, similarly to step S**104** in FIG. **4**A. That is, the notice control unit **206** causes the notifying unit **112** of the object vehicle to give the simplified caution notice to the user of the object vehicle by the auditory method. Thereby, the notice control unit **206** can perform a minimal caution by the auditory method, immediately after the inadequate driving operation, so as not to interfere with the driving of the user (driver) of the object vehicle that is likely to be in the traveling state.
(106) In step S**304**, the notice control unit **206** determines whether a predetermined time T**2** (for example, ten minutes) has elapsed since the occurrence of the accident or the near miss (that is, the occurrence of the inadequate driving operation). In the case where the predetermined time T**2** has elapsed, the notice control unit **206** proceeds to step S**306**, and in the case where the predetermined time T**2** has elapsed, the notice control unit **206** waits until the elapse of the predetermined time T**2** (that is, the notice control unit **206** repeats the process of step S**304**).
(107) The process of step S**304** may be skipped.
(108) In step S**306**, the notice control unit **206** determines whether the object vehicle is continuously at a standstill, based on the most recent vehicle information about the object vehicle that is stored in the vehicle information storage unit **202**, similarly to step S**106** in FIG. **4**A. In the case where the object vehicle is continuously at a standstill, the notice control unit **206** proceeds to step S**308**, and in the case where the object vehicle is not continuously at a standstill, the notice control unit **206** waits until the standstill state is continued (that is, the notice control unit **206** repeats the process of step S**306** until the standstill state is continued).
(109) In step S**308**, the notice control unit **206** sends, to the object vehicle, the caution information for performing the simplified caution to the user (driver) of the object vehicle by the visual method, similarly to step S**108** in FIG. **4**A. That is, the notice control unit **206** causes the notifying unit **112** of the object vehicle to give the simplified caution notice to the user of the object vehicle by the visual method. For example, the simplified caution notice screen **410** in FIG. **4**B is displayed on the display **15**B of the object vehicle, similarly to the above-described first example. Thereby, the user can recognize that there is a possibility that the user has performed the inadequate driving operation.
(110) In step S**310**, the notice control unit **206** determines whether the check operation for the detailed caution information has been performed on the simplified caution notice screen, similarly to step S**110** in FIG. **4**A. In the case where the check operation has been performed, the notice control unit **206** proceeds to step S**312**, and in the case where the check operation has not been performed, the notice control unit **206** proceeds to step S**314**.
(111) In step S**312**, the notice control unit **206** sends, to the object vehicle, the caution information for performing the detailed caution to the user (driver) of the object vehicle by the visual method, similarly to step S**112** in FIG. **4**A, and then ends this process. That is, the notice control unit **206** causes the notifying unit **112** of the object vehicle to give the detailed caution notice to the user of the object vehicle by the visual method. For example, the detailed caution notice screen **420** in FIG. **4**C is displayed on the display **15**B of the object vehicle, similarly to the above-described first example. Thereby, the notice control unit **206** can perform the detailed caution relevant to the inadequate driving operation, to the user of the object vehicle, at a timing when a certain amount of time has elapsed since the occurrence of the inadequate driving operation. Therefore, the user of the object vehicle can look back on the inadequate driving operation by the user, after the user settles down to some extent.
(112) Meanwhile, in step S**314**, the notice control unit **206** determines whether the IG-OFF of the object vehicle has been performed (that is, whether the object vehicle has been stopped), based on the most recent vehicle information about the object vehicle in the vehicle information storage unit **202**. In the case where the IG-OFF of the object vehicle has been performed, the notice control unit **206** proceeds to step S**316**, and in the case where the IG-OFF of the object vehicle has not been performed, the notice control unit **206** returns to step S**310**.

(113) In step S**316**, to the user terminal **40** that is used by the user of the object vehicle, the notice control unit **206** sends the caution information for performing the simplified caution to the user of the object vehicle by the visual method, similarly to step S**202** in FIG. **5**A. Thereby, the user terminal **40** (notifying unit **401**) that is used by the user of the object vehicle receives the caution information, and displays the simplified caution notice screen on the display device **46**. That is, the notice control unit **206** causes the notifying unit **401** of the user terminal **40** to give the detailed caution notice to the user of the object vehicle by the visual method. For example, the simplified caution notice screen **510** is displayed on the display device **46** of the user terminal **40** that is used by the user of the object vehicle, similarly to the above-described second example. Thereby, the user can recognize that there is a possibility that the user has performed the inadequate driving operation.
(114) In step S**318**, the notice control unit **206** determines whether the check operation for the detailed caution information has been performed on the simplified caution notice screen, similarly to step S**204** in FIG. **5**A. In the case where the check operation has been performed, the notice control unit **206** proceeds to step S**320**, and in the case where the check operation has not been performed, the notice control unit **206** proceeds to step S**322**.
(115) In step S**320**, to the user terminal **40** that is used by the user of the object vehicle, the notice control unit **206** sends the caution information for performing the detailed caution to the user of the object vehicle by the visual method, similarly to step S**206** in FIG. **5**A, and then ends this process. That is, the notice control unit **206** causes the notifying unit **401** of the user terminal **40** to give the detailed caution notice to the user of the object vehicle by the visual method. For example, the detailed caution notice screen **520** is displayed on the display device **46** of the user terminal **40**, similarly to the above-described second example. Thereby, in a situation where the user of the object vehicle has stopped the object vehicle (IG-OFF) without looking at the detailed caution notice screen through the display **15**B of the object vehicle, the notice control unit **206** can perform again the detailed caution relevant to the inadequate driving operation to the user of the object vehicle, through the user terminal **40**. Further, the notice control unit **206** performs again the detailed caution, only in the situation where the user of the object vehicle has stopped the object vehicle without looking at the detailed caution notice screen through the display **15**B of the object vehicle. Therefore, the notice control unit **206** can prevent the caution from being performed twice, and can prevent the user from having a feeling of discomfort.
(116) Meanwhile, in step S**322**, the notice control unit **206** determines whether the simplified caution notice has been hidden by a user's operation through the input device **47**, similarly to step S**208** in FIG. **5**A. In the case where the simplified caution notice has been hidden, the notice control unit **206** proceeds to step S**324**, and in the case where the simplified caution notice has not been hidden, the notice control unit **206** returns to step S**318**.
(117) In step S**324**, the notice control unit **206** determines whether a predetermined time T**3** (for example, ten minutes) has elapsed since the hiding of the simplified caution notice, similarly to step S**210** in FIG. **5**A. In the case where the predetermined time T**3** has not elapsed, the notice control unit **206** waits until the elapse of the predetermined time T**3** (that is, the notice control unit **206** repeats the process of S**324** until the elapse of the predetermined time T**3**). In the case where the predetermined time T**3** has elapsed, the notice control unit **206** returns to step S**316**, and displays the simplified caution notice screen on the display device **46** again. Thereby, even when the user has hidden the simplified caution notice without checking the detailed caution notice, the vehicle monitoring server **20** can display the simplified caution notice screen on the display device **46** again, and can provide an opportunity to perform the detailed caution relevant to the inadequate driving operation to the user of the object vehicle.
(118) Operation of Embodiment
(119) Next, the operation of the vehicle monitoring system **1** (vehicle monitoring server **20**) according to the embodiment will be described.

(120) In the embodiment, the accident/near miss detecting unit **203** detects a first vehicle **10** that is of the plurality of vehicles **10** and that is involved in the near miss or the accident, based on the predetermined vehicle information acquired by each of the plurality of vehicles **10** and including the position information and the date-and-time information (acquisition date-and-time information) corresponding to the time when the vehicle information is acquired by the plurality of vehicles **10**. In the case where the first vehicle **10** is detected by the accident/near miss detecting unit **203**, the surrounding vehicle extracting unit **204** extracts the surrounding vehicles that are of the plurality of vehicles **10** and that are positioned at the area surrounding the first vehicle **10** when the first vehicle **10** is involved in the near miss or the accident, based on the position information and acquisition date-and-time information about the plurality of vehicle **10**. Then, based on the vehicle information about the surrounding vehicles, the object vehicle specifying unit **205** specifies a second vehicle **10** that is of the surrounding vehicles and in which the inadequate driving operation causing the near miss or the accident of the first vehicle **10** has been performed.
(121) Thereby, the vehicle monitoring server **20** can know the motion state of the vehicle **10**, the driving operation state by the driver, and the like, from the vehicle information acquired by each vehicle **10**, and thereby, can detect the occurrence of the near miss (for example, a sudden braking) or accident (for example, a collision) of the vehicle **10**. Further, the vehicle monitoring server **20** can extract the surrounding vehicles positioned at the area surrounding the first vehicle **10** when the first vehicle **10** is involved in the near miss or the accident, from the position information (specifically, the position information about the first vehicle **10** and the position information about the other vehicles **10**) about each vehicle **10** and the date-and-time information corresponding to the time when the vehicle information is acquired. Therefore, the vehicle monitoring server **20** can know the motion state of the surrounding vehicles, the driving operation state by the driver, and the like, from the extracted vehicle information about the surrounding vehicles, and thereby, can specify the second vehicle **10** in which the inadequate driving operation corresponding to a cause of the near miss or accident of the first vehicle **10** has been performed, from the extracted surrounding vehicles.
(122) In the embodiment, the notice control unit **206** gives the notice of the caution to the user of the second vehicle **10** specified by the object vehicle specifying unit **205**, through the user terminal (for example, the navigation device **15** of the second vehicle **10** or the user terminal **40** that is used by the user of the second vehicle **10**).
(123) Thereby, the vehicle monitoring server **20** can perform the caution relevant to the inadequate driving operation causing the near miss or accident of the first vehicle **10**, to the user of the second vehicle **10** in which the inadequate driving operation has been performed.
(124) In the embodiment, in the case where the second vehicle **10** is traveling, the notice control unit **206** may give the auditory notice to the user that rides in the second vehicle **10**, through the in-vehicle device (navigation device **15**) of the second vehicle **10** or the mobile terminal (user terminal **40**) possessed by the user as the user terminal, and in the case where the second vehicle **10** is at a standstill, the notice control unit **206** may give the visual notice to the user that rides in the second vehicle **10**, through the in-vehicle device of the second vehicle **10** or the mobile terminal possessed by the user.
(125) Thereby, the vehicle monitoring server **20** can give the auditory notice to the user of the second vehicle during traveling, so as not to obstruct the driving operation, and can give the visual notice to the user of the second vehicle at a standstill, so as to perform a more specific caution.
(126) In the embodiment, in the case where the second vehicle **10** is in the stop state, the notice control unit **206** may give the notice of the caution to the user of the second vehicle **10**, through the user terminal (user terminal **40**) different from the in-vehicle device of the second vehicle **10**.
(127) Thereby, the vehicle monitoring server **20** can perform the caution relevant to the inadequate driving operation to the user of the second vehicle, at such a timing that the driving operation of the second vehicle is not obstructed.

(128) In the embodiment, the accident/near miss detecting unit **203** successively perform the detection of the first vehicle in response to the movement of the plurality of vehicles, and the surrounding vehicle extracting unit **204** and the object vehicle specifying unit **205** respectively perform the extraction of the surrounding vehicles and the specification of the second vehicle, immediately in response to the detection of the first vehicle by the accident/near miss detecting unit **203**. In response to the specification of the second vehicle by the object vehicle specifying unit **205**, the notice control unit **206** may give the auditory notice to the user that rides in the second vehicle **10**, through the in-vehicle device (navigation device **15**) of the second vehicle **10** or the mobile terminal (user terminal **40**) possessed by the user as the user terminal at a first timing that is a relatively early timing after the first vehicle is involved in the near miss or the accident, and then may give the visual notice to the user that rides in the second vehicle **10**, through the in-vehicle device of the second vehicle **10** or the mobile terminal possessed by the user, at a second timing that is a relatively late timing at which the second vehicle **10** is at a standstill.
(129) Thereby, the vehicle monitoring server **20** can perform a simplified caution to the user (driver) of the second vehicle, shortly after the accident or the near miss, even during traveling, and can perform a specific caution later, in the state where the second vehicle is at a standstill.
(130) In the embodiment, in the case where it is determined that the user is not looking at the content of the notice (detailed caution notice) at the second timing, the notice control unit **206** may give the notice of the caution to the user through the user terminal (user terminal **40**) different from the in-vehicle device of the second vehicle **10**, when the second vehicle **10** gets into the stop state.
(131) Thereby, even in the case where the user is not looking at the content of the notice of the caution during the driving of the second vehicle **10**, the vehicle monitoring server **20** can perform again the specific caution to the user of the second vehicle **10**, through the user terminal (for example, a smartphone) other than the in-vehicle device, after the stop of the second vehicle **10**.
(132) The embodiment of the disclosure has been described above in detail. The disclosure is not limited to the particular embodiment, and various modifications and improvements can be made within the scope of the spirit of the disclosure described in the claims.

Claims

1. An information processing device comprising: circuitry configured to: detect a first vehicle of a plurality of vehicles based on vehicle information, the first vehicle being involved in a near miss or an accident, the vehicle information being acquired by each of the plurality of vehicles and including, for each respective one of the plurality of vehicles, position state information, motion state information, operation state information, and control state information, wherein the near miss is an abrupt maneuver indicating avoidance or potential for a non-contact incident, and the accident is a collision or a rollover; extract information of surrounding vehicles of the plurality of vehicles based on the position information and date-and-time information, the date-and-time information corresponding to a time when the vehicle information is acquired by the plurality of vehicles, and the surrounding vehicles being positioned at an area surrounding the first vehicle when the first vehicle is involved in the near miss or the accident; specify a second vehicle of the surrounding vehicles based on the vehicle information about the surrounding vehicles, the second vehicle being a vehicle in which a driving operation causing the near miss or the accident of the first vehicle is performed; after the near miss or the accident of the first vehicle has occurred, give a first notice of a caution regarding the near miss or the accident to a user of the second vehicle through an in-vehicle device equipped in the second vehicle, the first notice of the caution including a simplified notice that omits at least one of a time of occurrence, location, or content of the driving operation causing the near miss or the accident; when a condition is satisfied, give a second notice of a caution regarding the near miss or the accident to the user of the second vehicle through the in-vehicle device after giving the first notice, the second notice of the caution including a detailed

notice including additional information on a cause of the near miss or the accident, the additional information including a time of occurrence, a location, or a type of driving operation; and in a case where the second vehicle is in a stop state, give the first notice of the caution to the user of the second vehicle through a user terminal, which is different from the in-vehicle device.

2. The information processing device according to claim 1, wherein the circuitry is further configured to: in a case where the second vehicle is traveling, give an auditory notice of the caution to the user of the second vehicle through the in-vehicle device, and in a case where the second vehicle is at a standstill, give a visual notice of the caution to the user through the in-vehicle device.

3. The information processing device according to claim 1, wherein the circuitry is further configured to: perform the detection of the first vehicle immediately in response to movement of the plurality of vehicles; perform the extraction of the surrounding vehicles and the specification of the second vehicle immediately in response to the detection of the first vehicle; and in response to the specification of the second vehicle: give an auditory notice of the caution to the user of the second vehicle through the in-vehicle device at a first timing that is a relatively early timing after the first vehicle is involved in the near miss or the accident, and then give a visual notice of the caution to the user of the second vehicle, through the in-vehicle device, at a second timing that is a relatively late timing relative to the relatively early timing at which the second vehicle is at a standstill.

4. The information processing device according to claim 3, wherein the circuitry is further configured to, in a case where it is determined that the user is not looking at a content of the visual notice at the second timing, give the visual notice to the user through a user terminal, which is different from the in-vehicle device, when the second vehicle gets into an ignition-OFF state.

5. The information processing device according to claim 1, wherein the circuitry is further configured to: give the first notice of the caution to the user through the in-vehicle device when the second vehicle is not in a stopped state, and give the first notice of the caution to the user through a mobile terminal possessed by the user when the second vehicle is in the stopped state.

6. The information processing device according to claim 5, wherein the stopped state is an ignition-OFF state.

7. The information processing device according to claim 6, wherein the circuitry is further configured to: give the simplified notice to the user through the in-vehicle device when the second vehicle is not in the stopped state and is traveling, and give the detailed notice to the user through the in-vehicle device when the second vehicle is not in the stopped state and is at a standstill.

8. The information processing device according to claim 1, wherein the circuitry is further configured to: give the simplified notice to the user through the in-vehicle device when the second vehicle is not in an ignition-OFF state and is traveling, and give the detailed notice to the user through the in-vehicle device when the second vehicle is not in the ignition-OFF state and is at a standstill.

9. The information processing device according to claim 1, wherein the circuitry is further configured to: determine whether the second vehicle is in a traveling state or a stopped state; give the first notice as an auditory notice during the traveling state; and give the second notice as a visual notice during the stopped state.

10. An information processing method that is executed by an information processing device, the information processing method comprising: detecting a first vehicle of a plurality of vehicles based on vehicle information, the first vehicle being involved in a near miss or an accident, the vehicle information being acquired by each of the plurality of vehicles and including, for each respective one of the plurality of vehicles, position state, information, motion state information, operation state information, and control state information, wherein the near miss is an abrupt maneuver indicating avoidance or potential for a non-contact incident, and the accident is a collision or a rollover; extracting information of surrounding vehicles of the plurality of vehicles based on the position information and date-and-time information corresponding to a time when the vehicle

information is acquired by the plurality of vehicles, the surrounding vehicles being positioned at an area surrounding the first vehicle when the first vehicle is involved in the near miss or the accident; specifying a second vehicle of the surrounding vehicles based on the vehicle information about the surrounding vehicles, the second vehicle being a vehicle in which a driving operation causing the near miss or the accident of the first vehicle is performed; after the near miss or the accident of the first vehicle has occurred, giving a first notice of a caution regarding the near miss or the accident to a user of the second vehicle through an in-vehicle device equipped in the second vehicle, the first notice of the caution including a simplified notice that omits at least one of a time of occurrence, location, or content of the driving operation causing the near miss or the accident; when a condition is satisfied, giving a second notice of a caution regarding the near miss or the accident to the user of the second vehicle through the in-vehicle device after giving the first notice, the second notice of the caution including a detailed notice including additional information on a cause of the near miss or the accident, the additional information including a time of occurrence, a location, or a type of driving operation; and in a case where the second vehicle is in a stop state, giving the first notice of the caution to the user of the second vehicle through a user terminal, which is different from the in-vehicle device.

11. A non-transitory computer readable medium storing a program that causes an information processing device to execute: detecting a first vehicle of a plurality of vehicles based on vehicle information, the first vehicle being involved in a near miss or an accident, the vehicle information being acquired by each of the plurality of vehicles and including, for each respective one of the plurality of vehicles, position state information, motion state information, operation state information, and control state information, wherein the near miss is an abrupt maneuver indicating avoidance or potential for a non-contact incident, and the accident is a collision or a rollover; extracting information of surrounding vehicles of the plurality of vehicles based on the position information and date-and-time information corresponding to a time when the vehicle information is acquired by the plurality of vehicles, the surrounding vehicles being positioned at an area surrounding the first vehicle when the first vehicle is involved in the near miss or the accident; specifying a second vehicle of the surrounding vehicles based on the vehicle information about the surrounding vehicles, the second vehicle being a vehicle in which a driving operation causing the near miss or the accident of the first vehicle is performed; after the near miss or the accident of the first vehicle has occurred, giving a first notice of a caution regarding the near miss or the accident to a user of the second vehicle through an in-vehicle device equipped in the second vehicle, the first notice of the caution including a simplified notice that omits at least one of a time of occurrence, location, or content of the driving operation causing the near miss or the accident; when a condition is satisfied, giving a second notice of a caution regarding the near miss or the accident to the user of the second vehicle through the in-vehicle device after giving the first notice, the second notice of the caution including a detailed notice including additional information on a cause of the near miss or the accident, the additional information including a time of occurrence, a location, or a type of driving operation; and in a case where the second vehicle is in a stop state, give the first notice of the caution to the user of the second vehicle through a user terminal, which is different from the in-vehicle device.

12. An information processing device comprising: circuitry configured to: detect a first vehicle of a plurality of vehicles based on vehicle information immediately in response to movement of the plurality of vehicles, the first vehicle being involved in a near miss or an accident, the vehicle information being acquired by each of the plurality of vehicles and including, for each respective one of the plurality of vehicles, position state information, motion state information, operation state information, and control state information, wherein the near miss is an abrupt maneuver indicating avoidance or potential for a non-contact incident, and the accident is a collision or a rollover; extract, immediately in response to the detection of the first vehicle, information of surrounding vehicles of the plurality of vehicles based on the position information and date-and-time

information, the date-and-time information corresponding to a time when the vehicle information is acquired by the plurality of vehicles, and the surrounding vehicles being positioned at an area surrounding the first vehicle when the first vehicle is involved in the near miss or the accident; specify, immediately in response to the detection of the first vehicle, a second vehicle of the surrounding vehicles based on the vehicle information about the surrounding vehicles, the second vehicle being a vehicle in which a driving operation causing the near miss or the accident of the first vehicle is performed; after the near miss or the accident of the first vehicle has occurred, give a first notice of a caution regarding the near miss or the accident to a user of the second vehicle through an in-vehicle device equipped in the second vehicle, the first notice of the caution including a simplified notice that omits at least one of a time of occurrence, location, or content of the driving operation causing the near miss or the accident; when a condition is satisfied, give a second notice of a caution regarding the near miss or the accident to the user of the second vehicle through the in-vehicle device after giving the first notice, the second notice of the caution including a detailed notice including additional information on a cause of the near miss or the accident, the additional information including a time of occurrence, a location, or a type of driving operation; in response to the specification of the second vehicle: give an auditory notice of the caution to the user of the second vehicle through the in-vehicle device at a first timing that is a relatively early timing after the first vehicle is involved in the near miss or the accident, and then give a visual notice of the caution to the user of the second vehicle, through the in-vehicle device, at a second timing that is a relatively late timing relative to the relatively early timing at which the second vehicle is at a standstill; and in a case where it is determined that the user is not looking at a content of the visual notice at the second timing, give the visual notice to the user through a user terminal, which is different from the in-vehicle device, when the second vehicle gets into an ignition-OFF state.